MAST32001 Computational Statistics - Lecture 12

Variational Inference

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Motivation

Where are we?

In the previous lectures, Bayesian inference required the posterior

$$p(\theta \mid y) \propto p(y \mid \theta)p(\theta).$$

MCMC approximates posterior expectations using simulation:

$$E\{h(\theta) \mid y\} \approx \frac{1}{M} \sum_{m=1}^{M} h(\theta^{(m)}).$$

Variational inference takes a different approach:

replace sampling by optimization over probability distributions.

Why variational inference?

Variational inference (VI) chooses an approximating distribution

$$q(\theta) \in \mathcal{Q}$$

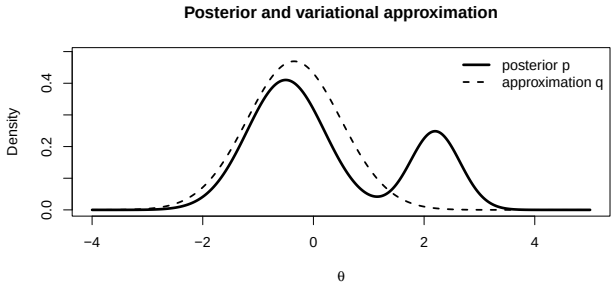
from a tractable family and makes it close to the posterior.

Potential advantages:

- often much faster than MCMC;
- deterministic optimization is possible in some models;
- scales well to large data and many latent variables (unobserved quantities introduced as part of a model);
- posterior summaries are immediately available from q.

But the result is an **approximation**, whose quality depends on $\mathcal{Q}$ and the optimization.

The basic picture



A simple family may be computationally convenient but unable to reproduce all posterior features.

KL divergence and the ELBO

Kullback–Leibler divergence

We write **KL** for the **Kullback–Leibler** divergence. For densities q and p,

$$\mathrm{KL}(q\|p) = \int q(\theta) \log \frac{q(\theta)}{p(\theta)}\, d\theta.$$

Equivalently,

$$\mathrm{KL}(q\|p) = E_q\{\log q(\theta) - \log p(\theta)\}.$$

Important properties:

- $\mathrm{KL}(q\|p) \geq 0$;
- it is zero iff $q = p$ almost everywhere;
- it is **not symmetric**;
- it is not a distance metric.

VI usually minimizes $\mathrm{KL}\{q(\theta)\|p(\theta \mid y)\}$.

Why not minimize KL directly?

Using Bayes' theorem,

$$\log p(\theta \mid y) = \log p(y, \theta) - \log p(y).$$

Therefore

$$\mathrm{KL}\{q \| p(\theta \mid y)\} = E_q\{\log q(\theta) - \log p(y, \theta)\} + \log p(y).$$

But

$$p(y) = \int p(y, \theta)\, d\theta$$

is exactly the difficult normalizing constant we wanted to avoid.

Evidence lower bound: ELBO

Rearranging gives

$$\log p(y) = \underbrace{E_q\{\log p(y,\theta)\} - E_q\{\log q(\theta)\}}_{\mathcal{L}(q)} + \mathrm{KL}\{q \| p(\theta \mid y)\}.$$

The marginal likelihood $p(y)$ is also called the **model evidence**. Define

$$\mathcal{L}(q) = E_q\{\log p(y,\theta)\} - E_q\{\log q(\theta)\}.$$

Thus $\mathcal{L}(q)$ is the **evidence lower bound (ELBO)**. Since KL is nonnegative,

$$\mathcal{L}(q) \leq \log p(y).$$

Thus

$$\boxed{\max_q \mathcal{L}(q) \iff \min_q \mathrm{KL}\{q \| p(\theta \mid y)\}.}$$

ELBO as fit minus complexity

The ELBO can also be written as

$$\mathcal{L}(q) = E_q\{\log p(y \mid \theta)\} - \mathrm{KL}\{q(\theta) \| p(\theta)\}.$$

It balances:

1. **fit to the observed data** through expected log likelihood;
2. **departure from the prior** through a KL penalty.

The optimizer is not simply maximizing the likelihood.

Jensen's inequality derivation

Start from

$$\log p(y) = \log \int q(\theta) \frac{p(y,\theta)}{q(\theta)}\, d\theta = \log E_q \left\{ \frac{p(y,\theta)}{q(\theta)} \right\}.$$

Because $\log$ is concave,

$$\log p(y) \geq E_q\{\log p(y,\theta) - \log q(\theta)\} = \mathcal{L}(q).$$

This gives the same lower bound without first introducing the posterior KL.

Reverse KL has characteristic behavior

VI commonly minimizes

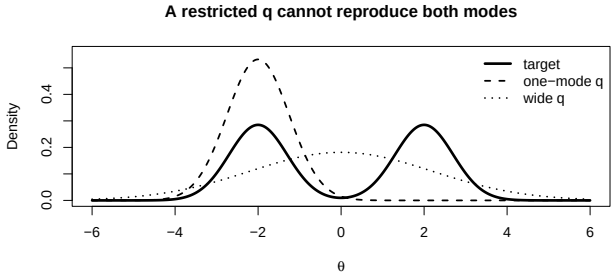
$$\mathrm{KL}(q\|p),$$

not $\mathrm{KL}(p\|q)$.

If q puts mass where p is almost zero, the penalty can be large. Reverse-KL VI can therefore be **mode-seeking** and under-dispersed.

For a multimodal target, a simple unimodal q may concentrate on one mode rather than cover all modes.

Visualization: multimodal target



Mean-field variational inference

Mean-field family

A common restriction is

$$q(\theta_1, \ldots, \theta_p) = \prod_{j=1}^{p} q_j(\theta_j).$$

This is the **mean-field** assumption.

It does **not** say that the posterior parameters are actually independent. We deliberately approximate the posterior by a distribution with independent components because this makes optimization easier.

This can be severe when the true posterior has strong dependence.

Coordinate ascent variational inference

Coordinate ascent variational inference (CAVI) updates one variational factor at a time.

Hold all factors except q_j fixed. The optimal coordinate update satisfies

$$\log q_j^*(\theta_j) = E_{q_{-j}}\{\log p(y, \theta)\} + \text{constant},$$

where

$$q_{-j}(\theta_{-j}) = \prod_{k \neq j} q_k(\theta_k).$$

Equivalently,

$$q_j^*(\theta_j) \propto \exp\left[E_{q_{-j}}\{\log p(y, \theta)\}\right].$$

Why the coordinate update works

For fixed q_{-j}, collect all terms involving q_j:

$$\mathcal{L}(q_j) = E_{q_j}\{f(\theta_j)\} - E_{q_j}\{\log q_j(\theta_j)\} + C,$$

where

$$f(\theta_j) = E_{q_{-j}}\{\log p(y, \theta)\}.$$

Define

$$\tilde{p}_j(\theta_j) \propto e^{f(\theta_j)}.$$

Then maximizing the ELBO is equivalent to minimizing

$$\mathrm{KL}(q_j \| \tilde{p}_j).$$

Hence $q_j^* = \tilde{p}_j$.

CAVI algorithm

Start with factors $q_1^{(0)}, \ldots, q_p^{(0)}$. Repeatedly update

$$q_1 \leftarrow q_1^*, \quad q_2 \leftarrow q_2^*, \quad \ldots, \quad q_p \leftarrow q_p^*.$$

Each exact coordinate update cannot decrease the ELBO.

CAVI resembles Gibbs structurally, but:

- Gibbs **samples** from full conditionals;
- CAVI **optimizes distributions** using expectations over other factors.

Interactive task 1

R TASK 1: KL between two normal distributions

Let

$$p = N(0,1), \qquad q = N(m, s^2).$$

Then

$$\mathrm{KL}(q\|p) = \frac{1}{2}\left(m^2 + s^2 - 1 - \log s^2\right).$$

In pairs, using R:

1. make a grid over $m \in [-2, 2]$ and $s \in [0.2, 2]$;
2. calculate KL on the grid;
3. find the minimizing (m, s);
4. draw a contour plot;
5. explain why the answer should be $(0, 1)$ without calculation.

R TASK 1: solution

```r
m_grid <- seq(-2, 2, length.out = 150)
s_grid <- seq(0.2, 2, length.out = 150)

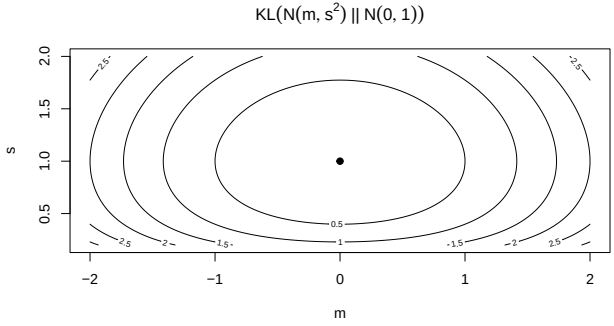
KL <- outer(
  m_grid, s_grid,
  Vectorize(function(m, s) {
    0.5 * (m^2 + s^2 - 1 - log(s^2))
  })
)

ind <- which(KL == min(KL), arr.ind = TRUE)[1, ]
c(m = m_grid[ind[1]], s = s_grid[ind[2]])
```

```
##           m           s
## -0.01342282  0.99731544
```

```r
contour(m_grid, s_grid, KL, xlab = "m", ylab = "s",
        main = expression(KL(N(m,s^2)~"||"~N(0,1))))
points(0, 1, pch = 19)
```

R TASK 1: solution



The minimum is attained at $q = p$, as required by the defining property of KL divergence.

Bayesian linear regression example

Model

Consider

$$y = X\beta + \varepsilon, \qquad \varepsilon \sim N_n(0, \sigma^2 I_n),$$

with known σ^2, and prior

$$\beta \sim N_p(0, \tau^2 I_p).$$

Then

$$p(\beta \mid y) \propto \exp\left\{ -\frac{1}{2\sigma^2}(y - X\beta)^\top (y - X\beta) - \frac{1}{2\tau^2}\beta^\top \beta \right\}.$$

The exact posterior is available, so we can directly assess VI.

Exact posterior

Collecting quadratic terms gives

$$\beta \mid y \sim N_p(m, V),$$

with

$$V = \left(\frac{X^\top X}{\sigma^2} + \frac{I_p}{\tau^2} \right)^{-1}, \qquad m = V \frac{X^\top y}{\sigma^2}.$$

Define

$$\Lambda = V^{-1} = \frac{X^\top X}{\sigma^2} + \frac{I_p}{\tau^2}.$$

Mean-field approximation

Choose

$$q(\beta) = \prod_{j=1}^{p} N(m_j, s_j^2).$$

The coordinate update is

$$s_j^2 = \left(\frac{x_j^\top x_j}{\sigma^2} + \frac{1}{\tau^2} \right)^{-1} = \frac{1}{\Lambda_{jj}},$$

and

$$m_j = \frac{s_j^2}{\sigma^2} x_j^\top \left(y - \sum_{k \neq j} x_k m_k \right).$$

Interpretation of the update

For coordinate j, form the partial residual

$$r_j = y - \sum_{k \neq j} x_k m_k.$$

Then

$$m_j = \frac{s_j^2}{\sigma^2} x_j^\top r_j.$$

Each update resembles a one-variable regression on the residual left after accounting for the other current coefficient means.

The variational variance is $s_j^2 = 1/\Lambda_{jj}$: it uses only a diagonal element of the posterior precision.

ELBO for the linear-model example

For

$$q(\beta) = \prod_{j=1}^{p} N(m_j, s_j^2),$$

the ELBO is the expected log likelihood plus the expected log prior plus the entropy of q.

```
elbo_lm <- function(X, y, m, s2, sigma2, tau2) {
  n <- nrow(X)
  p <- ncol(X)

  e_loglik <- -0.5 * n * log(2 * pi * sigma2) -
    0.5 / sigma2 * (
      sum((y - as.vector(X %*% m))^2) +
      sum(s2 * colSums(X^2))
    )

  e_logprior <- -0.5 * p * log(2 * pi * tau2) -
    0.5 / tau2 * sum(m^2 + s2)

  entropy_q <- 0.5 * sum(log(2 * pi * exp(1) * s2))

  e_loglik + e_logprior + entropy_q
}
```

CAVI in R

```r
cavi_lm <- function(X, y, sigma2, tau2, tol = 1e-10, max_iter = 10000) {
  p <- ncol(X)
  m <- rep(0, p)
  s2 <- 1 / (colSums(X^2) / sigma2 + 1 / tau2)
  elbo <- numeric(max_iter + 1)
  elbo[1] <- elbo_lm(X, y, m, s2, sigma2, tau2)

  for (iter in seq_len(max_iter)) {
    m_old <- m
    for (j in seq_len(p)) {
      r_j <- y - as.vector(X[, -j, drop = FALSE] %*% m[-j])
      m[j] <- s2[j] * as.numeric(crossprod(X[, j], r_j)) / sigma2
    }
    elbo[iter + 1] <- elbo_lm(X, y, m, s2, sigma2, tau2)
    if (max(abs(m - m_old)) < tol) break
  }
  list(mean = m, var = s2, iterations = iter, elbo = elbo[seq_len(iter + 1)])
}
```

Simulate correlated predictors

```r
set.seed(1201)
n <- 100
x1 <- rnorm(n)
x2 <- 0.95 * x1 + sqrt(1 - 0.95^2) * rnorm(n)
X <- cbind(x1, x2)
beta_true <- c(1, 1)
sigma2 <- 1
tau2 <- 100
y <- as.vector(X %*% beta_true + rnorm(n, sd = sqrt(sigma2)))
fit_vi <- cavi_lm(X, y, sigma2, tau2)
fit_vi[c("mean", "var", "iterations")]
```

```
## $mean
## [1] 0.7793964 0.9592087
##
## $var
##         x1         x2
## 0.01157257 0.01153367
##
## $iterations
## [1] 181
```

Compare with exact posterior

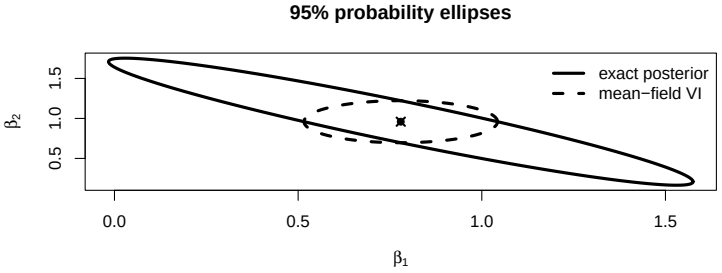
```r
p <- ncol(X)
Lambda <- crossprod(X) / sigma2 + diag(1 / tau2, p)
V_exact <- solve(Lambda)
m_exact <- as.vector(V_exact %*% crossprod(X, y) / sigma2)

rbind(
  exact_mean = m_exact,
  VI_mean = fit_vi$mean,
  exact_sd = sqrt(diag(V_exact)),
  VI_sd = sqrt(fit_vi$var)
)
```

```
##                   x1        x2
## exact_mean 0.7793964 0.9592087
## VI_mean    0.7793964 0.9592087
## exact_sd   0.3250362 0.3244894
## VI_sd      0.1075759 0.1073949
```

For this Gaussian target, mean-field CAVI recovers the posterior mean very well, but the marginal posterior variances need not be correct.

Exact posterior versus mean-field VI

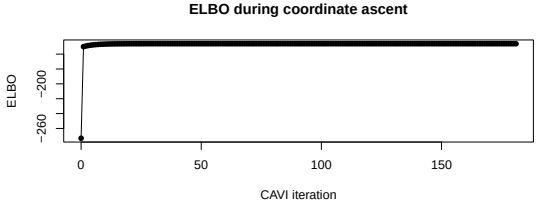


The exact posterior has a tilted ellipse because the coefficients are dependent. Mean-field VI forces an axis-aligned ellipse and cannot reproduce this dependence.

Inspecting the ELBO during CAVI

Each exact CAVI coordinate update cannot decrease the ELBO. Therefore the ELBO should increase and then level off as the optimization converges.

```
plot(seq_along(fit_vi$elbo) - 1, fit_vi$elbo, type = "o", pch = 16,
     xlab = "CAVI iteration", ylab = "ELBO",
     main = "ELBO during coordinate ascent")
```



A flat ELBO indicates that the **optimization has stabilized**. It does **not** show that the variational family is a good approximation to the true posterior.

For exact CAVI the ELBO is nondecreasing. With stochastic-gradient VI, an estimated ELBO will usually be noisy, so we look for an overall plateau rather than strict monotonicity.

Why are VI marginal variances too small?

For the exact Gaussian posterior,

$$\mathrm{Var}(\beta_j \mid y) = (\Lambda^{-1})_{jj}.$$

Mean-field VI gives

$$s_j^2 = \frac{1}{\Lambda_{jj}}.$$

In general,

$$(\Lambda^{-1})_{jj} \neq \frac{1}{\Lambda_{jj}}.$$

With correlated parameters, the difference can be substantial:

a good variational mean does not imply good uncertainty quantification.

Interactive task 2

R TASK 2: when does mean-field improve?

Use the regression code from the previous slides. Change the predictor correlation to

$$\rho \in \{0, 0.5, 0.95\}.$$

For each value:

1. run `cavi_lm()`;
2. compute the exact posterior covariance;
3. compare exact and VI posterior SDs;
4. explain what changes as ρ increases.

Predict the result **before** running the code.

R TASK 2: solution

```r
compare_rho <- function(rho) {
  set.seed(1203)
  x1 <- rnorm(n)
  x2 <- rho * x1 + sqrt(1 - rho^2) * rnorm(n)
  Xr <- cbind(x1, x2)
  yr <- as.vector(Xr %*% beta_true + rnorm(n))

  vi <- cavi_lm(Xr, yr, sigma2 = 1, tau2 = 100)
  Lambda <- crossprod(Xr) + diag(1 / 100, 2)
  V <- solve(Lambda)

  c(rho = rho,
    exact_sd1 = sqrt(V[1, 1]), VI_sd1 = sqrt(vi$var[1]),
    exact_sd2 = sqrt(V[2, 2]), VI_sd2 = sqrt(vi$var[2]))
}
```

R TASK 2: solution

```
res <- t(vapply(c(0, 0.5, 0.95), compare_rho, numeric(5)))
round(res, 3)
```

```
##       rho exact_sd1 VI_sd1.x1 exact_sd2 VI_sd2.x2
## [1,] 0.00     0.103       0.1     0.115     0.113
## [2,] 0.50     0.109       0.1     0.133     0.122
## [3,] 0.95     0.343       0.1     0.369     0.108
```

As posterior dependence increases, the independence restriction becomes more costly and mean-field marginal SDs become increasingly too small.

Beyond conjugate CAVI

What if updates are not analytic?

In many models,

$$E_{q_{-j}}\{\log p(y, \theta)\}$$

does not lead to a recognizable distribution.

Then parameterize $q_\lambda(\theta)$ and optimize

$$\mathcal{L}(\lambda) = E_{q_\lambda}\{\log p(y, \theta) - \log q_\lambda(\theta)\}$$

numerically.

This leads to **black-box variational inference** and stochastic-gradient methods.

Monte Carlo estimate of the ELBO

If

$$\theta^{(1)}, \ldots, \theta^{(S)} \sim q_\lambda,$$

then

$$\widehat{\mathcal{L}}(\lambda) = \frac{1}{S} \sum_{s=1}^{S} [\log p(y, \theta^{(s)}) - \log q_\lambda(\theta^{(s)})].$$

We can optimize this noisy objective using stochastic-gradient methods, but differentiating through random draws requires care.

Reparameterization trick

For

$$\theta \sim N(\mu, \sigma^2)$$

write

$$\varepsilon \sim N(0, 1), \qquad \theta = \mu + \sigma\varepsilon.$$

Now randomness is in ε, whose distribution does not depend on (μ, σ). This allows gradients with respect to variational parameters to be estimated by differentiating through the deterministic transformation.

Optimization diagnostics

Unlike MCMC, VI has no stationary-chain trace plot.

Useful checks include:

- ELBO versus iteration;
- sensitivity to initialization;
- multiple optimization runs;
- gradient size / optimizer convergence;
- posterior predictive checks;
- comparison with MCMC on smaller problems.

A stable ELBO says the optimizer stabilized for the chosen family; it does **not** prove that q is close to the posterior.

A frequentist viewpoint

VI is not exclusively Bayesian

A **latent variable** is an unobserved quantity introduced as part of a model. For example, in a mixture model the component label of each observation is latent.

Suppose a model contains observed data y, latent variables z, and parameter θ:

$$p(y \mid \theta) = \int p(y, z \mid \theta)\, dz.$$

For any $q(z)$,

$$\log p(y \mid \theta) \geq E_q\{\log p(y, z \mid \theta)\} - E_q\{\log q(z)\}.$$

The bound can be optimized over both q and the frequentist parameter θ.

EM as a variational algorithm

Preview: the expectation–maximization (EM) algorithm will also be studied next week. Here the goal is only to introduce it and show its connection with variational ideas.

Define

$$\mathcal{F}(q, \theta) = E_q\{\log p(y, z \mid \theta)\} - E_q\{\log q(z)\}.$$

Then

$$\log p(y \mid \theta) = \mathcal{F}(q, \theta) + \mathrm{KL}\{q(z) \| p(z \mid y, \theta)\}.$$

E-step:

$$q(z) \leftarrow p(z \mid y, \theta^{(t)}).$$

M-step:

$$\theta^{(t+1)} = \arg\max_{\theta} \mathcal{F}(q, \theta).$$

Thus classical EM is a special variational optimization scheme.

Gaussian mixture example

Suppose

$$Y_i \sim \pi N(\mu_1, \sigma^2) + (1-\pi)N(\mu_2, \sigma^2),$$

with latent $Z_i \in \{1, 2\}$.

The observed-data likelihood contains sums:

$$L(\theta) = \prod_{i=1}^{n} [\pi\phi(y_i; \mu_1, \sigma) + (1-\pi)\phi(y_i; \mu_2, \sigma)].$$

The latent-variable representation makes the complete-data likelihood much simpler.

E-step: responsibilities

For current parameters,

$$r_{i1} = P(Z_i = 1 \mid y_i) = \frac{\pi\phi(y_i; \mu_1, \sigma)}{\pi\phi(y_i; \mu_1, \sigma) + (1 - \pi)\phi(y_i; \mu_2, \sigma)}.$$

Similarly,

$$r_{i2} = 1 - r_{i1}.$$

Here the variational E-step is exact.

M-step

The expected complete-data log likelihood yields

$$\pi^{new} = \frac{1}{n}\sum_i r_{i1},$$

$$\mu_1^{new} = \frac{\sum_i r_{i1} y_i}{\sum_i r_{i1}}, \qquad \mu_2^{new} = \frac{\sum_i r_{i2} y_i}{\sum_i r_{i2}},$$

and

$$(\sigma^2)^{new} = \frac{1}{n}\sum_i [r_{i1}(y_i - \mu_1)^2 + r_{i2}(y_i - \mu_2)^2].$$

EM for a Gaussian mixture in R

```r
em_gmm <- function(y, max_iter = 500, tol = 1e-8) {
  n <- length(y)
  pi1 <- 0.5
  mu1 <- as.numeric(quantile(y, 0.25))
  mu2 <- as.numeric(quantile(y, 0.75))
  sigma <- sd(y)
  loglik <- numeric(max_iter)

  for (iter in seq_len(max_iter)) {
    a <- pi1 * dnorm(y, mu1, sigma)
    b <- (1 - pi1) * dnorm(y, mu2, sigma)
    r1 <- a / (a + b)

    pi1 <- mean(r1)
    mu1 <- sum(r1 * y) / sum(r1)
    mu2 <- sum((1 - r1) * y) / sum(1 - r1)
    sigma <- sqrt(mean(r1 * (y - mu1)^2 +
                       (1 - r1) * (y - mu2)^2))

    loglik[iter] <- sum(log(
      pi1 * dnorm(y, mu1, sigma) +
      (1 - pi1) * dnorm(y, mu2, sigma)
    ))
    if (iter > 1 &&
```

Run the mixture example

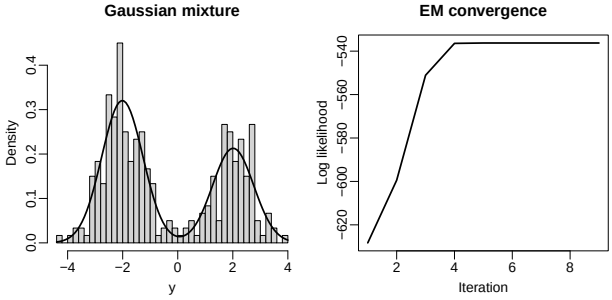
```r
set.seed(1204)
y_mix <- c(rnorm(180, -2, 0.8), rnorm(120, 2, 0.8))
fit_em <- em_gmm(y_mix)
fit_em[c("pi", "mu", "sigma", "iterations")]
```

```
## $pi
## [1] 0.6006798
##
## $mu
## [1] -2.012786  2.012095
##
## $sigma
## [1] 0.7479764
##
## $iterations
## [1] 9
```

Run the mixture example

```r
par(mfrow = c(1, 2), mar = c(3, 3, 3, 0.7), mgp = c(1.7, 0.5, 0))
hist(y_mix, probability = TRUE, breaks = 30,
     main = "Gaussian mixture", xlab = "y")
curve(fit_em$pi * dnorm(x, fit_em$mu[1], fit_em$sigma) +
      (1 - fit_em$pi) * dnorm(x, fit_em$mu[2], fit_em$sigma),
      add = TRUE, lwd = 2)
plot(fit_em$loglik, type = "l", lwd = 2,
     xlab = "Iteration", ylab = "Log likelihood",
     main = "EM convergence")
par(mfrow = c(1, 1))
```

Run the mixture example



Bayesian VI versus frequentist variational estimation

	Bayesian VI	Frequentist / latent-variable view
Unknowns	θ has posterior	θ is a parameter
Approximate	$p(\theta \mid y)$	latent distribution / bound
Optimize	$q(\theta)$	$q(z)$ and θ
Objective	posterior ELBO	likelihood lower bound
Example	mean-field Bayes	EM / variational EM

The mathematics is closely related, but the inferential interpretation differs.

Frequentist uncertainty requires care

A variational distribution $q(\theta)$ is **not automatically** a frequentist sampling distribution for an estimator.

For frequentist inference one may need separate theory for:

- consistency;
- asymptotic distribution;
- standard errors;
- confidence intervals.

Do not interpret a convenient variational variance as a universal uncertainty measure.

Practical VI

RStan ADVI: define a complete example

RStan's `vb()` needs a compiled `stanmodel`. Here is a complete normal-model example that can be copied into an R session.

```
library(rstan)

normal_code <- "
data {
  int<lower=1> N;
  vector[N] y;
}
parameters {
  real mu;
  real<lower=0> sigma;
}
model {
  mu ~ normal(0, 10);
  sigma ~ exponential(1);
  y ~ normal(mu, sigma);
}
"

y_stan <- c(-0.4, 0.2, 0.7, 1.1, 0.5, -0.1)
stan_data <- list(N = length(y_stan), y = y_stan)
```

RStan ADVI: compile and fit

`stan_model()` compiles the model. The resulting `stanmodel` can be passed to `vb()`.

```
normal_stan <- stan_model(model_code = normal_code)

fit_vb <- vb(
  object = normal_stan,
  data = stan_data,
  seed = 20260812
)

print(fit_vb, pars = c("mu", "sigma"))
```

By default, `vb()` uses a mean-field Gaussian approximation; `algorithm = "fullrank"` uses a Gaussian approximation with a full covariance matrix.

RStan currently describes `vb()` as experimental and recommends `sampling()`/`stan()` for final inference when reliable posterior accuracy is required.

MCMC or VI?

MCMC aims for asymptotically exact posterior simulation, subject to convergence and Monte Carlo error.

VI solves an optimization problem inside a restricted family.

VI is attractive when:

- model or data are large;
- approximate inference is sufficient;
- repeated fitting is required;
- MCMC is prohibitively expensive.

MCMC is attractive when accurate uncertainty quantification and posterior shape are important and computation is feasible.

Common failure modes

VI can fail because:

- $\mathcal{Q}$ is too restrictive;
- mean-field removes important dependence;
- reverse KL misses modes or tails;
- the optimizer reaches a poor local optimum;
- gradients are noisy;
- parameterization is poor;
- a converged ELBO hides approximation error.

Separate:

Did the optimization converge?

from

Is the approximation good enough?

What should we check?

At least:

1. inspect ELBO / optimization history;
2. use several starting values where possible;
3. inspect marginal and joint posterior shapes;
4. perform posterior predictive checks;
5. compare important summaries with MCMC when feasible;
6. ask whether the family can represent known dependencies or multimodality;
7. be cautious with variational SDs and tail probabilities.

Summary

Main ideas

Variational inference turns inference into optimization:

$$q^* = \arg\min_{q \in \mathcal{Q}} \mathrm{KL}\{q \| p(\theta \mid y)\}.$$

Equivalently, maximize

$$\mathcal{L}(q) = E_q\{\log p(y, \theta)\} - E_q\{\log q(\theta)\}.$$

Mean-field VI gives tractable coordinate updates but may underestimate dependence and uncertainty.

Modern VI uses stochastic gradients and reparameterization.

The same variational principle appears in frequentist latent-variable methods, with EM as an important special case.

Take-home message

MCMC asks

Can we construct a chain that samples from the posterior?

VI asks

Can we find a tractable distribution close enough to the posterior?

VI can be much faster, but introduces a new source of error:

approximation error from the variational family and optimization.